Robert William **Moerman**

South African · PhD in Physics Candidate · Preferred Name: Robert Moerman-Redding
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Education

Doctor of Philosophy (PhD) in Mathematical Physics — *Hatfield, England*
University of Hertfordshire — *October 2019 - present*

- Supervisor: Dr Tomasz Łukowski (t.lukowski@herts.ac.uk)
- Dissertation: "Positive Geometries for Scattering Amplitudes in Momentum Space: How Positivity Breathes Combinatorial Life"
- Publications: [1–8]

Master of Advanced Study in Applied Mathematics — *Cambridge, England*
University of Cambridge — *October 2018 - June 2019*

A one-year Masters-level taught degree, awarded with *distinction* on 19 July 2019. Average Grade: 89%.

- Examination Ranking: 14 out of 256 candidates
- Courses: Quantum Field Theory 85% • Symmetries, Fields & Particles 85% • Statistical Field Theory 91% • Advanced Quantum Field Theory 95% • String Theory 86% • General Relativity 92%

Master of Science (MSc) in Theoretical Physics — *Cape Town, South Africa*
University of Cape Town (UCT) — *January 2016 - April 2018*

A two-year Masters-level research degree by dissertation [9], awarded with *distinction* on 3 April 2018.

- Supervisor: A/Prof. Heribert Weigert (heribert.weigert@uct.ac.za)
- Dissertation: "A gauge-invariant, symmetry-preserving truncation of JIMWLK"

Bachelor of Science Honours (BSc Hons.) in Physics — *Cape Town, South Africa*
University of Cape Town (UCT) — *January 2015 - December 2015*

A one-year Honours-level degree by course work and minor dissertation, awarded in the *first class* on 23 December 2015. Average Grade: 92%.

- Courses: Electrodynamics 1 • Electrodynamics 2 • Statistical Physics • Quantum Mechanics 1 • Quantum Mechanics 2 • Relativistic Quantum Mechanics • String Theory • Classical Mechanics • Differential Geometry • Advanced Mathematical Methods 2
- Supervisor: A/Prof. William Horowitz (wa.horowitz@uct.ac.za)
- Project: "A semi-classical recipe for wobbly-limp noodles in partonic soup" which culminated in [10, 11].

Bachelor of Science (BSc) with majors in Applied Mathematics and Physics — *Cape Town, South Africa*
University of Cape Town (UCT) — *January 2012 - December 2014*

A three-year undergraduate degree, awarded with *distinction in Applied Mathematics and Physics*, and the degree with *distinction* on 15 December 2014. Average Grade: 96%.

Licences & Certifications

DeepLearning.AI TensorFlow Developer — *Coursera*
Credential ID: D2B9XPN29DFV — *July 2022*

Sequences, Time Series and Prediction *Coursera*
CREDENTIAL ID: V8FSU5AYNKXG *July 2022*

Natural Language Processing in TensorFlow *Coursera*
CREDENTIAL ID: FMPG7R8659B9 *July 2022*

Convolutional Neural Networks in TensorFlow *Coursera*
CREDENTIAL ID: ZL65TXRQVCXX *July 2022*

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning *Coursera*
CREDENTIAL ID: MK9VLED987HS *July 2022*

Macine Learning *Coursera*
CREDENTIAL ID: BR693GKYR97C *June 2022*

Recent Presentations

Amplitudes Workshop *Oxford, England*
MATHEMATICAL INSTITUTE, UNIVERSITY OF OXFORD *6 July 2022*
Invited talk "Pushing Forward Rational Differential Forms".
(https://inspirehep.net/seminars/2128054)

Positive Geometries in Scattering Amplitudes and Beyond *Mainz, Germany*
MAINZ INSTITUTE FOR THEORETICAL PHYSICS, JOHANNES GUTENBERG UNIVERSITY *7-18 June 2021*
Invited talk "Kleiss-Kuijf Relations from Momentum Amplituhedron Geometry" via Zoom.
(https://inspirehep.net/seminars/1925864)

Novel Ideas in HEP: From Theory to Detector *Southampton, England*
UNIVERSITY OF SOUTHAMPTON *13-14 April 2021*
Student talk "Scattering Amplitudes from Positive Geometries" via Microsoft Teams.
(https://inspirehep.net/seminars/1925865)

Organisational Roles

Geometry and Scattering Amplitudes Journal Club
CO-FOUNDED WITH DAVID DAMGAARD *October 2020 - present*
A journal club via Zoom aimed at PhD students and early-career postdocs across the globe working at the intersection of geometry and scattering amplitudes, meeting fortnightly.
(https://lists.physik.uni-muenchen.de/mailman/listinfo/geometryandscatteringamplitudes)

Honours & Awards

SCHOLARSHIPS

2019-2023	**Research Studentship**, from the University of Hertfordshire, Hatfield	*England*
2018-2019	**Bursary for Students from Africa**, from Trinity College, Cambridge	*England*
2016-2017	**Masters Bursary**, from the National Institute for Theoretical Physics (NITheP)	*South Africa*
2014-2015	**Undergraduate and Honours Bursary**, from the Square Kilometer Array South Africa (SKA)	*South Africa*

Academic Awards

2019	**Senior Scholar**, awarded for having completed the MAST degree with a First	*Trinity College, Cambridge*
2016	**Myer Levinson (Emdin) Scholarship**, awarded every second year to a candidate who has obtained the BSc Hons. degree in the first class and who proposes to pursue further study.	*UCT, South Africa*
2014	**Ivor Levin Memorial Scholarship**, awarded to the best student in second-year Physics, registered in the Faculty of Science, who will be proceeding to third-year Physics.	*UCT, South Africa*
2013	**Twamley December Scholarship**, awarded for the most outstanding academic performance at the end of the first year of study.	*UCT, South Africa*
2012	**The Mathematics & Applied Mathematics Webb-Ellis trophy**, awarded to the best student in first year with double majors in Applied Mathematics and Mathematics.	*UCT, South Africa*
2012	**Science Entrance Scholarship**, awarded for excellent performance in the National Senior Certificate (NSC) examinations.	*UCT, South Africa*

Teaching

Visiting Lecturer *Hatfield, England*
University of Hertfordshire *February 2020 - present*
I have been a class tutor for undergraduate and honours-level modules, including set theory, differential equations, complex analysis and quantum field theory.

Tutor *Cape Town, South Africa*
University of Cape Town *2015 - 2018*
I facilitated first- and second-year undergraduate physics computational labs and tutorials. I also tutored third-year undergraduate atomic physics and honours-level electodynamics.

Publications and Preprints

PhD

[1] Tomasz Lukowski, **Robert Moerman**, and Jonah Stalknecht. "Pushforwards via Scattering Equations with Applications to Positive Geometries". In: (June 2022). Submitted to JHEP. arXiv: 2206.14196 [hep-th].

[2] **Robert Moerman** and Lauren K. Williams. "Grass trees and forests: Enumeration of Grassmannian trees and forests, with applications to the momentum amplituhedron". In: (Dec. 2021). Submitted to Combinatorial Theory. arXiv: 2112.02061 [math.CO].

[3] Tomasz Łukowski, **Robert Moerman**, and Jonah Stalknecht. "On the geometry of the orthogonal momentum amplituhedron". In: (Dec. 2021). Submitted to JHEP. arXiv: 2112.03294 [hep-th].

[4] Livia Ferro and **Robert Moerman**. "The Grassmannian for celestial superamplitudes". In: *JHEP* 11 (2021), p. 187. DOI: 10.1007/JHEP11(2021)187. arXiv: 2107.07496 [hep-th].

[5] David Damgaard, Livia Ferro, Tomasz Łukowski, and **Robert Moerman**. "Kleiss-Kuijf Relations from Momentum Amplituhedron Geometry". In: *JHEP* 07 (2021), p. 111. DOI: 10.1007/JHEP07(2021)111. arXiv: 2103.13908 [hep-th].

[6] David Damgaard, Livia Ferro, Tomasz Łukowski, and **Robert Moerman**. "Momentum Amplituhedron meets Kinematic Associahedron". In: *JHEP* 02 (2021), p. 041. DOI: 10.1007/JHEP02(2021)041. arXiv: 2010.15858 [hep-th].

[7] Livia Ferro, Tomasz Łukowski, and **Robert Moerman**. "From Momentum Amplituhedron Boundaries to Amplitude Singularities and Back". In: *JHEP* 07.07 (2020), p. 201. DOI: 10.1007/JHEP07(2020)201. arXiv: 2003.13704 [hep-th].

[8] Tomasz Łukowski and **Robert Moerman**. "Boundaries of the amplituhedron with amplituhedron-Boundaries". In: *Comput. Phys. Commun.* 259 (2021), p. 107653. DOI: 10.1016/j.cpc.2020.107653. arXiv: 2002.07146 [hep-th].

BSc Hons

[10] A. K. Mes, **Robert Moerman**, J. P. Shock, and W. A. Horowitz. "Strongly Coupled Heavy and Light Quark Thermal Motion from AdS/CFT". In: *Annals of Physics* (2021), p. 168675. ISSN: 0003-4916. DOI: 10.1016/j.aop.2021.168675. arXiv: 2008.09196 [hep-th].

[11] **Robert Moerman** and W. A. Horowitz. "A semi-classical recipe for wobbly limp noodles in partonic soup". In: (May 2016). arXiv: 1605.09285 [hep-th].

Theses

MSc

[9] **Robert Moerman**. *A gauge-invariant, symmetry-preserving truncation of JIMWLK*. MSc Thesis. University of Cape Town. 2018. URL: https://open.uct.ac.za/bitstream/handle/11427/27998/thesis_sci_2018_moerman_robertwilliam.pdf?sequence=1.